Full Title: "Urban density and the loneliness epidemic: Developing and validating a tool to assess the role of housing type and design in social connection/isolation"

Summary

Lay Title: Density and the loneliness epidemic: Developing a tool to assess housing impacts

Lay Abstract

1 BACKGROUND & OBJECTIVES

Roughly one in five Canadians report experiencing social isolation (20%) and loneliness (19%),[1] while nearly 60% indicate experiencing a lack of community connection, of which 16% report that they feel 'no sense of community at all.'[2] Of those experiencing social isolation, close to half (49%) say they feel their health is either 'fair' or 'poor,' while those reporting feeling frequently lonely rate their overall life satisfaction lower when compared to those who says they felt lonely less frequently.[3] Presently in 2025, *social isolation,* the absence of meaningful and supportive relationships with other individuals,[4] and *loneliness*, the "awareness of being fundamentally separated from other people and from the universe ... [via feelings of] sadness, hopelessness, grief, meaninglessness or anguish,"[5(p1322)] are a policy priorities in Canada.[e.g., 6,7] By some accounts, isolation and are an expanding "epidemic" that threatens health and society,[8,9] and one that in Canada is particularly consequential for sex/gender minority, disabled, and older populations, as well as those living with long-term conditions.[10] Collectively, the associated health care costs of loneliness and isolation are significant as these conditions are leading contributors to Canadians' poor mental, which the Canadian Mental Health Association estimates costs $50 billion annually in direct health care, lost productivity, and lost quality of life.[11]

Canadians spend almost 90% of their daily time indoors, the considerable majority of which is at home (69.9%).[12] While 35.9% of time at home is spent sleeping,[12] the centrality of the home environment with respect to health, well-being, early life development, and later life aging is increasingly well documented as a leading social determinant of health (SDoH) across the life course.[13,14] For instance, the condition of housing (i.e., good condition v. need minor repairs v. need major repairs) can have disparate impacts on mental and physical health outcomes for residents,[15] while the level of artificial lighting in the home can negatively affect indicators such as sleep health.[16] Beyond physical and mental health implications, it is increasingly understood that housing can play a key role in social well-being, with reported links being noted to sense of community belonging,[17] sense of pride,[18] and social/community integration.[19]

In Canada, the national housing stock has diversified across low-, mid-, and high-density models in recent years[20] in response to urban sustainability measures (e.g., limiting sprawl), concerns pertaining to the loss of productive agricultural land,[21] and the promotion of more compact and accessible economies of scale.[22] However, despite the growing evidence of housing as a consequential SDoH with links to social health and well-being, and Canada's diversifying housing mix, little is currently known about what distinct built, natural, and social designs of the home environment affect social health and well-being (i.e., depth, quality of one's relationships with others, their neighbourhood, and their community[23]) as existing evidence has ostensibly only tracked health and well-being indicators across housing types (e.g., single family home, apartment) in the limited reviewed comparative studies.[24] Moreover, there is a dearth of extant evidence-supported housing strategies to support health, and especially those that consider achieving key metrics of health and equity.[25] The aim of this research is to support effective evidence-based planning and public health practice via the development of a novel, interdisciplinary, expert informed, and psychometrically validated 'pro-social' housing design tool for practitioners and researchers. This research seeks to achieve three/four objectives:

Objective 1 (O1): Examine which social, built, and natural structures and features of the home environment among low- (i.e., 1-2 storey buildings; e.g., single family home), medium- (i.e., 3-8 storey buildings; e.g., condos, townhomes), and high-rise (i.e., >9 storey buildings; e.g., high-rise apartments) dwelling residents (N=1,500) perceive as most important to their social health.

Objective 2 (O2): Conduct a comparative multi-city (i.e., N=15/Waterloo, N=15/Edmonton, N=15/Montreal) photovoice study investigating resident perspectives of low-, medium-, and high-density housing regarding pro-/anti-social environmental design in/around their home.

Objective 3a (O3a): Triangulate the findings of O1-O2 via expert audits to develop the 'socially connective housing/home environment design – universal evaluation' (SCHEDule) tool to evaluate the social design quality of different housing environments.

Objective 3b (O3b): Validate the SCHEDule tool with community surveys featuring visuals of different housing models and design quality as identified in the O3.1 expert audits.

2 RESEARCH METHODS & TIMELINES

2.1 Network Analysis of Home Env. Influences of Social Health in Low-, Mid-, & High-Rise Dwellings *(O1: May/26-Feb/27)*: What social (e.g., friendly neighbours, cultural designs/artifacts, security cameras), built (e.g., courtyards, wide sidewalks, narrow adjacent streets), and natural (e.g., property soft edges, tall trees, gardens) environment features do low-, mid-, and high-rise dwelling residents connect the most to their perceived sense of community, social well-being, sense of belonging, sense of security, and life satisfaction?

Data Collection: ...

Data Analysis: social network analysis (SNA) of a sample of high-rise dwellers (i.e., 18+ years living in a building ≥8 storeys; N=X,XXX) in the 10 largest municipalities in Canada.

In the SNA, nodes will be individuals (i.e., CCHS participant IDs) … edges will be social provisions (i.e.,) …

Analyses will control for age, sex, ethnicity, socioeconomic status (i.e., education + income), and size of building (# of storeys).

2.2 Multi-City Photovoice Study of Social Health and Housing Design *(O2: Mar/27-Sept/27)*: A three-city (i.e., RoW, XXX, XXX) ...

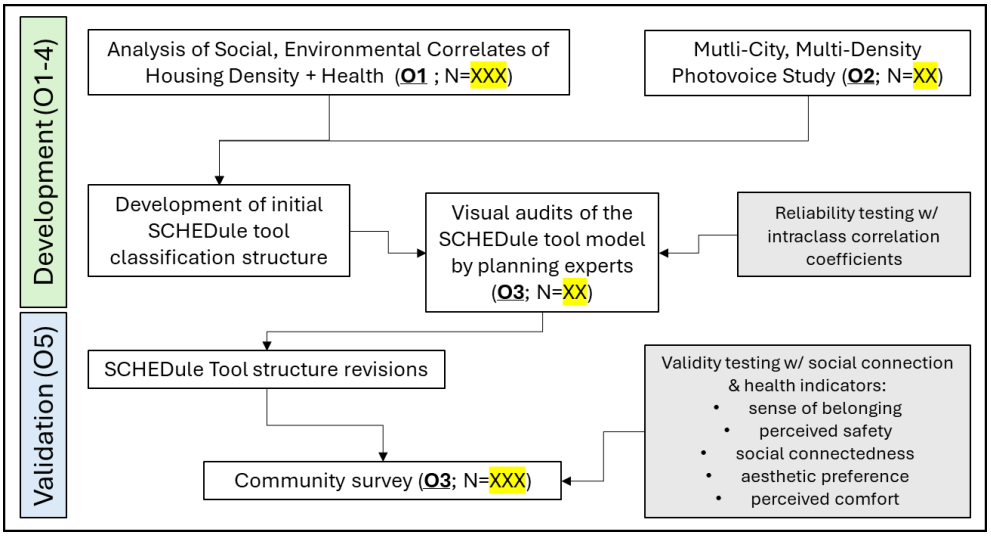
Recruitment: Purposive sampling[26] focused on recruiting participants living in low- (i.e., single family home), medium- (i.e., townhomes, duplexes, mid-rise apartments/condos 2-8 storeys), and high-density (i.e., 9 or more storey apartments/condos) housing environments will be conducted in each city resulting in a collective sample of 45 (15 per site).

Data Collection: Photovoice

Data Analysis: Thematic

2.3 Development of the 'SCHEDule' tool *(O3a: Oct/27-Apr/28)*: Synthesizing the findings from the O1-O3 studies will be used to create a list of assessment items reflecting pro-social and anti-social housing designs. For example, measures of pro-social designs may include aesthetics (e.g., natural amenities or spaces,[27] attractive housing model or historical architecture[28]), active travel infrastructure,[29] safety/crime prevention design,[30] and visibility, while anti-social measures may feature physical hazards (e.g., narrow sidewalks,[31] hard edge property designs[32]), social disconnection/lack of cleanliness (e.g., few fourth [informal] spaces[33]), motorized traffic/parking infrastructure,[28] and tall fences/property dividers. The list of items will be the initial model of the 'socially connective housing/home environment design – universal evaluation' (SCHEDule) tool which will be assessed for reliability (O4) and validity (O5) (see analytical plan in Figure X).

Figure X. Flow Chart for the Development and Validation of the SCHEDule Tool



Recruitment: Independent raters who will test the tool will be recruited as urban planning and design professionals (N=15-20). In addition to being topic experts, we will aim to recruit a sample with an average career length of at least 5 years. Experts will be invited to complete an online survey wherein they will be instructed to assess the social connectivity/pro-social design quality—according to the developed list of items—of 45 photos featuring low-, medium-, and high-density housing environments (15 images per density). Participants will be given the initial SCHEDule tool featuring the initial pro- and anti-social design categories with descriptions of each item. Participants will be instructed to do their best to avoid rating the quality of image, weather conditions in the image, or their personal preferences about the content of the image.

Housing Environment Images: To ensure the contemporary validity of the analysis, 45 images will be collected from the O3 study to represent real world examples of the pro- and anti-social design themes generated from the O1-O3 findings (i.e., anti-social low/medium/high-density, pro-social low/medium/high-density). Thus, 15 images representing housing in three densities from varied municipalities will be assessed by each independent rater. Raters will be instructed to evaluate each image for its social connectivity quality across the developed categories on a 6-point Likert (1: Strongly Disagree, 2: Disagree, 3: Slightly Disagree, 4: Slightly Agree, 5: Agree, 6: Strongly Agree). The rating scale will not feature a 'neutral' option in order to mitigate acquiescence bias—evaluator reluctance to respond with more extreme options which can contribute to central tendency biases that result in systematic error during the evaluation.[34]

Reliability Analysis: Intraclass coefficients (ICC) tests will be calculated first to measure the homogeneity between experts when rating housing environments. ICCs indicate whether, and the extent to which, the scores given by raters are correlated[35] (~~i.e., to which extent their ratings of the social connectivity of the housing environment increase/decrease in the same direction from one context to the next~~). ICC models will be considered as raters being picked randomly from the population of topic experts (random effect), and the reliability measures being based on the profile of their ratings. Thus, a Two-Way Random model will be used to evaluate the consistency of rankings (i.e., ICC, 95% Confidence Intervals) across both single items and averages per rater. Any items with an ICC < 0.50[35] will be dropped before final evaluations of the tool structure.

Assessments of the reliability of the structure of the initial SCHEDule tool will then be completed via exploratory factor analyses (EFA) and computing McDonald's Omegas (ω). First, a latent variable modelling for a two-factor SCHEDule tool (pro-social, anti-social housing designs) will be run to collect standardized loadings (λ) and error variances (θ).[36] Second, ω coefficients will be computed using the formula $\omega = (\Sigma\lambda_i)^2 / [(\Sigma\lambda_i)^2 + \Sigma\theta_i]$, where: λ_i = standardized factor loading for item *I* and θ_i = error variance for item *i*. Cutoffs for factor loadings will be $\omega \geq 0.70$-0.80 (good-to-excellent) reliability[37] to finalize the structure of the tool. McDonald's Omegas are widely recognized as one of the most trustworthy estimates of reliability.[38] Omegas have been favourably argued against more traditional Cronbach's alpha (α) due to having fewer assumptions (e.g., unlike α, ω doesn't require tau-equivalence [equal factor loadings]) and more analytical flexibility (i.e., α assumes a unidimensional factor [one factor] and can underestimate reliability in multidimensional scales, while ω is grounded in latent variable modeling which is more powerful for multidimensional scales/tools).[39,40] All analyses will be performed in SPSS and the AMOS package.

2.4 Validation of the 'SCHEDule' tool *(O3b: Jan/28-Apr/28)*: Using anonymized visuals (i.e., all address and street name info blurred/removed) of low-, medium, and high-density environments from the O3 study, visual assess will be conducted by members of the public (N=XXX) ...

Data Collection: online survey (N=XXX)

Validity Analysis: perceived stress questionnaire ,

Confirmatory factor analysis (CFA) ...

Predictive validity analysis ... linear regressions

3 CHALLENGES & MITIGATION PLANS

Limited relevance to rural communities ...

4 RESEARCH TEAM

Our interdisciplinary team comprises strategically chosen experts who work closely with knowledge users to help lay the foundation for future urban population health intervention research (PHIR). **NPA Minaker** will oversee all aspects of the project. Minaker is an Associate Professor in Canada's largest urban planning school and FCI Director. The proposed research builds directly on her research program exploring health and urban design.[31,41–43]

Co-PA Cooke is a demographer working in the area of public health and social inequities. He has experience using Statistics Canada data and linked datasets including multiple years of CCHS.

Co-A Perrott is an ECR and Registered Professional Planner with expertise in urban design for public spaces in densifying cities. She will supervise Co-op1 and co-lead the KMb activities given her role as Associate Director, Professional Practice at Waterloo's School of Planning.

Co-A Glover ...

Co-A ... in 'Berta

Co-A ... in Monty

Co-A Buttazzoni is a postdoctoral fellow (PDF) at Waterloo. This research builds on his PHIR,[44–46] methodological and psychometric tool development,[47–49] and his emerging research on universal design to support health.[50,51]

Collaborator **Happy Cities** is an urban planning, research, and engagement firm with extensive experience in health and social well-being and serve in an advisory capacity on KMb activities. See *Participant Table* for more details.

PKU ...

5 KNOWLEDGE MOBILIZATION

6 ANTICIPATED IMPACTS

This will be the first national study in Canada to explore how high-rise location, urban greenness, and ALE are associated with social isolation/connectedness in a national sample of Canadians ≥18 years. Our intersectional SGBA+ approach will help identify intersectional identities for whom urban design interventions might be most effective. By leveraging FCI support and our multi-sector city-building networks, our findings will directly inform realistic, equity-focused municipal policy recommendations for Canadian municipalities.

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